# SURAJ JAYBHAYE

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## EDUCATION

**COEP Technological University (COEP Tech)** *Pune, India*
*Bachelor's Degree in Computer Science Engineering* *2020 – 2024*

## EXPERIENCE

**Invade Code Ltd.** *Kharadi, Pune*
*Associate FullStack Developer* *Jan 2026 – Present*

- Owned end-to-end project and feature delivery — from requirement gathering to deployment — for a multi-tenant Agri ERP platform (Next.js, TypeScript, Firebase, Firestore) serving **20+ branches and processing 1000+ daily document/record updates** across retail, inventory, and finance-linked workflows.
- Refactored the frontend and backend into modular, reusable RESTful components to **reduce duplicate logic and improve long-term maintainability**, and debugged and resolved production issues including data-fetch failures, edge-case crashes, and cross-feature linking bugs.
- Built an end-to-end n8n automation pipeline that monitors incoming Gmail messages via keyword filtering, auto-generates a job listing, and publishes it directly to LinkedIn — **cutting client posting time from 20–30 minutes to under a minute** with zero manual steps.
- Collaborated with cross-functional team members on feature delivery and mentored interns on Git workflows and development/deployment practices.

**Nebula Technology** *Pune*
*Software Engineer Intern* *Sep 2024 – Feb 2025*

- Developed full-stack features using React.js, Node.js, Express.js, and MySQL in an Agile environment.
- Built RESTful APIs, optimized SQL queries, resolved application issues, and participated in Git-based code reviews and testing.

## PROJECTS

**GovIntel | AI-Powered Government Policy Intelligence Platform**
*Next.js, TypeScript, Python/FastAPI, Qdrant, Elasticsearch, PostgreSQL, OCR, LLM (Gemini/OpenAI), RAG*

- Built an AI-powered RAG platform for Kerala government officers to search, summarize, and draft against hundreds of G.O.s, circulars, acts, and rules across departments, **cutting policy lookup time from ~15 minutes of manual search to under 30 seconds** with cited, hallucination-free answers in English and Malayalam.
- Designed a hybrid retrieval pipeline combining Qdrant vector search (multilingual-e5-small embeddings) for semantic/synonym matching with Elasticsearch BM25 for exact-term and G.O.-number lookup, merged via Reciprocal Rank Fusion, plus an automated OCR ingestion pipeline that auto-extracts title, department, and G.O. number from uploaded PDFs.

**UPI Offline Mesh — Mesh-Routed Deferred Payment Settlement | GitHub**
*Java 17, Spring Boot, JPA/Hibernate, H2, RSA-OAEP/AES-256-GCM*

- Designed and built a Spring Boot backend with a software-simulated Bluetooth mesh demonstrating offline payment propagation — encrypting a payment on the sender's device, gossiping it through untrusted intermediary nodes, and settling it once a bridge node regains connectivity.
- Implemented hybrid RSA-OAEP + AES-256-GCM encryption (the same authenticated-encryption scheme used by TLS) to keep payloads opaque and tamper-evident across untrusted devices, paired with a two-layer replay defense (24-hour signed freshness window + per-packet nonce) to reject stale or replayed packets before they reach the ledger.
- Engineered exactly-once settlement under concurrent duplicate delivery using atomic compare-and-set on the ciphertext's SHA-256 hash, backed by a unique DB index as a defense-in-depth fallback; validated with JUnit concurrency tests firing 3 simultaneous duplicate deliveries and **asserting exactly one settlement, two duplicate-drops, and a single sender debit**.

## TECHNICAL SKILLS

**Languages:** Java, Python, SQL, JavaScript (ES6+), TypeScript

**Frameworks & Libraries:** Spring Boot, React.js, Next.js, Node.js, RESTful Web Services, DSA

**Databases & Cloud:** PostgreSQL, MySQL, Redis, Firebase, Firestore, AWS, Docker

**AI/ML:** LLM (Gemini, OpenAI, Groq/Llama 3.3), RAG, Qdrant, Elasticsearch, Agentic Software Development

**Tools & OS:** Linux, Git, GitHub, Postman, n8n

## ACHIEVEMENTS

- Ranked **103rd** in MHT-CET among **540,000+ candidates** in Maharashtra State.
- Secured **1st Rank** in Ahilyanagar District in the 12th HSC Board Examination.